Architectural Inductive Bias and Visual Recognition: A Comparative Evaluation of MLP, CNN, and RNN-Based Feature Learning

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https://github.com/Zarenawa/architectural-inductive-bias-study

Abstract

Architectural inductive bias plays a fundamental role in determining how neural networks learn and represent image data. This study investigates its influence on image classification performance through a comparative analysis of three neural network architectures: a Convolutional Neural Network (CNN), a Multi-Layer Perceptron (MLP), and a Recurrent Neural Network (RNN). All models were trained and evaluated on a five-class flower image dataset using identical preprocessing, data augmentation, optimization settings, and training protocols to ensure a fair comparison. Model performance was assessed using Top-1 Accuracy, Macro Precision, Macro Recall, Macro F1-score, and Macro AUC, together with qualitative analyses based on confusion matrices, t-SNE visualizations, and occlusion sensitivity maps. The experimental results showed that the CNN consistently achieved the best overall performance, obtaining a Top-1 Accuracy of 68.67% and a Macro AUC of 91.28%, while the MLP and RNN achieved comparable but lower classification performance. The qualitative analyses further demonstrated that the CNN learned more discriminative feature representations and relied on more informative image regions during prediction. These findings support the hypothesis that architectural inductive bias significantly influences image classification performance, with convolution-based architectures providing a clear advantage for visual recognition tasks over architectures designed primarily for general vector processing or sequential data modeling.

***K*eywords** Architectural inductive bias · image classification · CNN · MLP · RNN

1 Introduction

Image classification is one of the fundamental tasks in computer vision, aiming to automatically assign images to predefined categories. It has been widely applied in various domains, including medical diagnosis, agricultural monitoring, autonomous driving, industrial inspection, and intelligent surveillance. [1], [2], [3], [4] Recent advances in deep learning have significantly improved image classification performance by enabling neural networks to learn hierarchical feature representations directly from raw image data. As a result, deep neural networks have become the dominant approach for solving image classification problems. [5] [6], [7] Among deep learning architectures, Convolutional Neural Networks (CNNs) have achieved remarkable success because their convolutional operations effectively capture local spatial patterns and hierarchical visual features. [8] [6] [7] In contrast, Multi-Layer Perceptrons (MLPs) process images as flattened feature vectors without explicitly preserving spatial relationships, while Recurrent Neural Networks (RNNs), originally designed for sequential data, can be adapted to image classification by interpreting images as sequences. [9], [10], [11] These architectural differences introduce distinct inductive biases, which influence how each model learns and represents visual information[12]. Although CNNs are generally recognized as the preferred architecture for image classification, comparatively fewer studies have examined CNNs, MLPs, and RNNs under a unified experimental framework where the dataset, preprocessing, training protocol, and evaluation metrics are kept consistent[13]. Such controlled comparisons are valuable because they help isolate the effect of architectural design rather than differences arising from data preparation or optimization strategies [14]. Therefore, this study investigates the influence of architectural inductive bias on image classification performance through a comparative analysis of CNN, MLP, and RNN models using a five-class flower classification dataset. All models were trained under identical experimental conditions, including the same preprocessing pipeline, data augmentation strategy, optimizer settings, and evaluation protocol. Performance was assessed using both quantitative metrics, including Top-1 Accuracy, Macro Precision, Macro Recall, Macro F1-score, and Macro AUC, and qualitative analyses based on confusion matrices, t-SNE visualizations, and occlusion sensitivity maps. By combining these complementary evaluation methods, this study aims to provide a comprehensive understanding of how architectural inductive bias affects image classification performance.

2 Related Work

2.1 Convolutional Neural Networks for Image Classification

Deep learning has become the dominant approach for image classification, with Convolutional Neural Networks (CNNs) demonstrating outstanding performance across a wide range of computer vision tasks [13]. Early architectures such as LeNet [15] established the foundation for convolution-based image recognition, while later models, including AlexNet [6], VGG [16], ResNet [7], and EfficientNet [17], significantly improved classification accuracy through deeper and more efficient network designs. The success of CNNs is largely attributed to their ability to exploit local spatial correlations and hierarchical feature representations in images.

2.2 Multi-Layer Perceptrons for Image Classification

Multi-Layer Perceptrons (MLPs) were among the earliest neural network architectures applied to classification problems [18]. Traditional MLPs process images as flattened vectors and therefore do not explicitly preserve spatial information. Although recent MLP-based vision architectures have demonstrated competitive performance by incorporating specialized design strategies, conventional MLPs generally remain less effective than CNNs for image classification because they lack an inherent spatial inductive bias [19].

2.3 Recurrent Neural Networks for Image Classification

Recurrent Neural Networks (RNNs), including Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) models, were originally developed for sequential data such as text, speech, and time-series analysis [13]. Nevertheless, several studies have adapted RNNs for image classification by representing images as sequences of rows, columns, or extracted feature vectors. While these approaches demonstrate that RNNs can perform image classification, their sequential processing mechanism is generally less suitable than convolutional operations for capturing two-dimensional spatial structures [11], [13].

2.4 Research Gap and Motivation

Previous comparative studies have primarily focused on maximizing classification accuracy using advanced architectures or comparing models under different experimental settings. However, relatively few studies have performed controlled comparisons of CNNs, MLPs, and RNNs using identical datasets, preprocessing procedures, training configurations, and evaluation protocols. Such comparisons are valuable because they isolate the influence of architectural inductive bias rather than differences introduced by training strategies or hyperparameter selection.

Therefore, this study conducts a controlled comparative analysis of CNN, MLP, and RNN models on a five-class flower image classification dataset. By maintaining consistent preprocessing, data augmentation, optimization settings, and evaluation criteria across all models, the study aims to provide a fair assessment of how architectural inductive bias influences image classification performance.

3 Methodology

3.1 Research Objective

The objective of this study is to investigate the effect of architectural inductive bias on image classification performance by comparing three fundamentally different neural network architectures: a Multi-Layer Perceptron (MLP), a Convolutional Neural Network (CNN), and a Recurrent Neural Network (RNN). The comparison is conducted on the same five-class flower classification dataset under a unified experimental protocol, where all models are trained using identical data splits, image preprocessing procedures, augmentation strategy, optimization settings, and evaluation metrics. By controlling these experimental conditions, the study aims to determine whether differences in classification performance can be primarily attributed to the inherent architectural characteristics of the models rather than variations in the training procedure.

3.2 Dataset

The experiments were conducted using a five-class flower image classification dataset consisting of the following categories: Daisy, Dandelion, Rose, Sunflower, and Tulip [20]. The dataset contains natural flower photographs collected under diverse environmental conditions, including variations in viewpoint, lighting, background, scale, and occlusion, making the classification task representative of real-world image recognition scenarios.

To evaluate the three neural network architectures under identical conditions, the dataset was divided into separate training, validation, and test sets while preserving samples from each flower category in every split. The resulting dataset distribution is summarized in Table 1.

Table 1: Dataset distribution across training, validation, and test sets.

Class	**Training**	**Validation**	**Test**
Daisy	350	75	76
Dandelion	452	97	97
Rose	347	75	75
Sunflower	346	74	75
Tulip	424	91	92
Total	**1919**	**412**	**415**

3.3 Data Preprocessing and Augmentation

To ensure a fair comparison among the three neural network architectures, all input images were processed using an identical preprocessing pipeline. First, each image was resized and center-cropped to a fixed resolution of 128×128 pixels, providing a consistent input size for all models. The images were then converted to tensors and normalized using the ImageNet mean and standard deviation values (mean = [0.485, 0.456, 0.406], standard deviation = [0.229, 0.224, 0.225]).

During training, a light data augmentation strategy was applied to improve the diversity of the training samples while preserving the semantic content of the images. The augmentation pipeline consisted of resizing the image to 1.14 times the target resolution, performing a center crop to 128×128 pixels, and applying a random horizontal flip with a probability of 0.3. No augmentation was applied to the validation or test sets [21, **?**]. Instead, these datasets underwent only deterministic resizing, center cropping, tensor conversion, and normalization to ensure that model evaluation remained consistent and unbiased across all experiments.

3.4 Network Architectures

Three neural network architectures, namely a Convolutional Neural Network (CNN), a Multi-Layer Perceptron (MLP), and a Recurrent Neural Network (RNN), were implemented to investigate the influence of architectural inductive bias on image classification. To ensure a fair comparison, all models were trained using the same dataset, preprocessing pipeline, optimization strategy, and training configuration. The primary difference among the models lies in their architectural design and the way they process image information.

3.4.1 Convolutional Neural Network (CNN)

The CNN extracts local spatial features using convolutional layers with shared kernels. These features are progressively refined through successive convolutional operations before being aggregated and passed to fully connected layers for classification. The use of local receptive fields and weight sharing provides the CNN with a strong spatial inductive bias, making it highly effective for image recognition tasks.

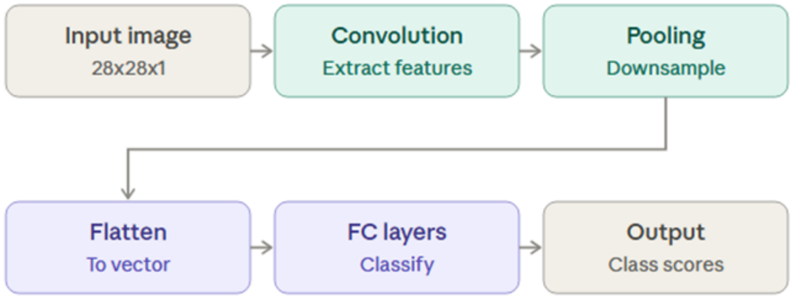


Figure 1: Architecture of the Convolutional Neural Network (CNN).

3.4.2 Multi-Layer Perceptron (MLP)

The MLP processes images by first flattening them into one-dimensional feature vectors, which are then passed through multiple fully connected layers. Since the spatial relationships between neighboring pixels are not explicitly preserved, the MLP must learn these relationships entirely from the training data. Consequently, it serves as a useful baseline model with relatively weak architectural inductive bias for image classification.

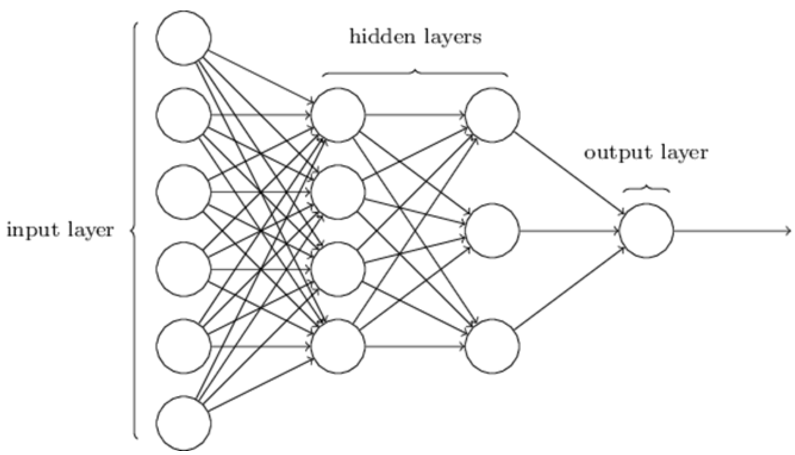


Figure 2: Architecture of the Multi-Layer Perceptron (MLP).

3.4.3 Recurrent Neural Network (RNN)

The RNN treats an image as a sequence by processing rows of pixels through recurrent units before generating the final classification. This sequential processing enables the model to capture dependencies across the input sequence but is less effective at modeling two-dimensional spatial structures than convolutional operations. Therefore, the RNN provides an alternative architectural inductive bias for comparison with the CNN and MLP in this study.

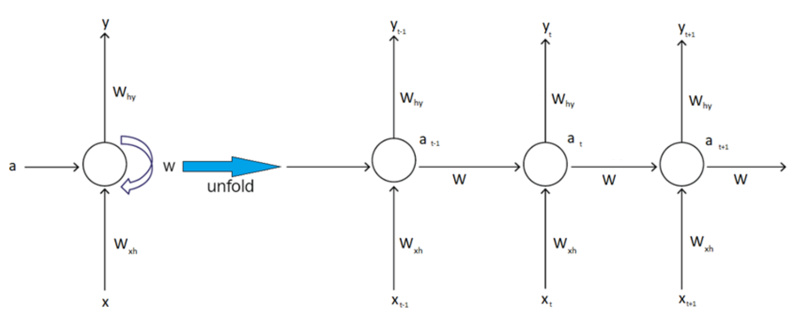


Figure 3: Architecture of the Recurrent Neural Network (RNN).

3.5 Training Configuration

To ensure a fair comparison among the three neural network architectures, all experiments were conducted using identical training settings. The models were trained under the same optimization strategy, learning rate, regularization techniques, and stopping criteria so that performance differences could be attributed primarily to architectural inductive bias rather than variations in hyperparameter selection. The complete training configuration is summarized in Table 2.

Table 2: Training configuration used for all experiments.

Parameter	**Value**
Dataset	Flowers
Number of classes	5
Input image size	128×128
Batch size	64
Maximum epochs	40
Optimizer	AdamW
Learning rate	0.001
Weight decay	0.0005
Adam β_1	0.9
Adam β_2	0.999
Adam ϵ	1×10^{-8}
Label smoothing	0.1
Dropout rate	0.3
Data augmentation	Light
Early stopping	Patience = 6

The AdamW optimizer was selected because it provides stable optimization while decoupling weight decay from gradient updates, thereby improving model generalization. Label smoothing and dropout were employed to reduce overfitting, whereas early stopping monitored validation performance to prevent unnecessary training after convergence. Applying the same training configuration across all models ensured that the experimental comparison remained consistent and unbiased.

3.6 Evaluation Metrics

3.7 Evaluation Metrics

The performance of the CNN, MLP, and RNN models was evaluated using both quantitative metrics and qualitative visualization techniques. Quantitative evaluation included Top-1 Accuracy, Macro Precision, Macro Recall, Macro F1-score, and Macro Area Under the Receiver Operating Characteristic Curve (Macro AUC), providing a comprehensive assessment of overall classification performance and class-wise discrimination. Macro-averaged metrics were selected to ensure that each flower class contributed equally to the evaluation, regardless of class frequency.

In addition to the quantitative metrics, several visualization techniques were employed to provide further insight into model behavior. Confusion matrices were used to analyze class-specific prediction errors, while Receiver Operating Characteristic (ROC) curves and Precision–Recall (PR) curves were used to evaluate classification performance across different decision thresholds. Furthermore, t-SNE visualizations were employed to examine the separability of learned feature representations, and occlusion sensitivity maps were generated to identify the image regions that contributed most significantly to the models' predictions [22], [23]. Together, these quantitative and qualitative analyses provide a comprehensive evaluation of the influence of architectural inductive bias on image classification performance.

4 Experimental Results

4.1 Overall Performance Comparison

The overall classification performance of the CNN, MLP, and RNN models was evaluated on the flower image classification dataset using Top-1 Accuracy, Macro Precision, Macro Recall, Macro F1-score, and Macro Area Under the Receiver Operating Characteristic Curve (Macro AUC). These metrics provide a comprehensive assessment of overall classification accuracy, balanced class-wise performance, and discriminative capability. The quantitative comparison of the three architectures is presented in Table 3.

Table 3: Overall performance comparison of CNN, MLP, and RNN.

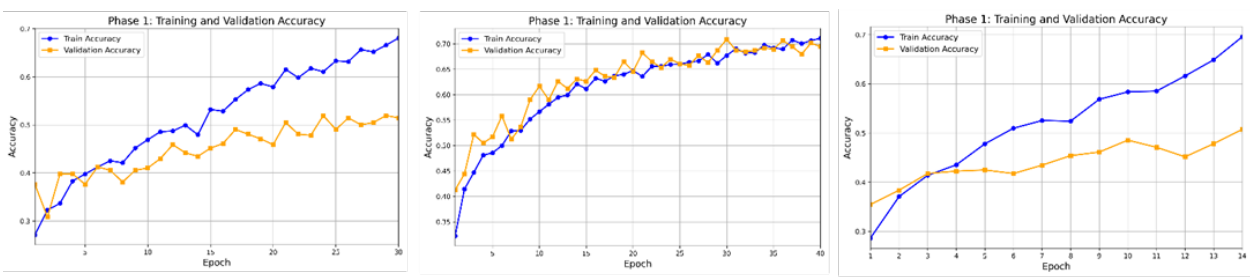
Model	Top-1 Accuracy (%)	Macro Precision (%)	Macro Recall (%)	Macro F1 (%)	Macro AUC (%)
MLP	44.09	45.31	43.88	44.09	77.63
CNN	**68.67**	**68.84**	**68.32**	**67.98**	**91.28**
RNN	45.06	44.93	44.90	43.88	77.86

As shown in Table 3, the CNN consistently achieved the highest performance across all evaluation metrics, obtaining a Top-1 Accuracy of 68.67% and a Macro AUC of 91.28%. These results indicate that the convolutional architecture was more effective at learning discriminative visual features than either the MLP or the RNN.

The MLP and RNN produced comparable results, with Top-1 Accuracy values of 44.09% and 45.06%, respectively, demonstrating substantially lower performance than the CNN. Although the RNN slightly outperformed the MLP in Top-1 Accuracy and Macro AUC, the differences between the two architectures were relatively small. Overall, the experimental results suggest that architectures incorporating spatial inductive bias provide a significant advantage for image classification, whereas architectures designed primarily for vector processing or sequential data modeling are less effective for this task.

4.2 Training Performance

The training and validation curves of the MLP, CNN, and RNN models are presented in Figure 4. These curves illustrate the learning process of each architecture throughout training and provide insight into convergence behavior and generalization performance.



(a) MLP (b) CNN (c) RNN

Figure 4: Training and validation curves of the evaluated architectures.

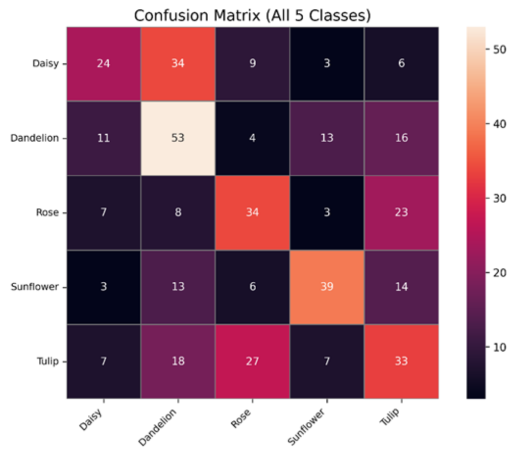
As shown in Figure 4, all three models exhibited a steady decrease in training and validation loss during the early stages of training before gradually converging. Early stopping was employed to terminate training when the validation loss no longer improved, thereby reducing unnecessary computation and helping to prevent overfitting.

Among the three architectures, the CNN demonstrated the fastest and most stable convergence while achieving the highest validation accuracy. The relatively small gap between its training and validation curves suggests good generalization to unseen data. In contrast, the MLP and RNN converged more slowly and stabilized at lower validation accuracies, indicating comparatively weaker learning performance under the same training conditions. These observations are consistent with the quantitative results presented in the previous section and further support the effectiveness of convolutional inductive bias for image classification.

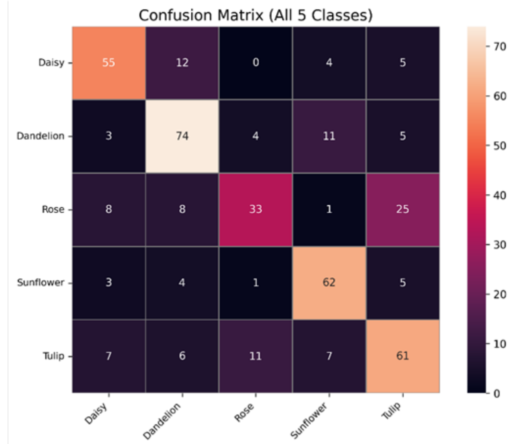
4.3 Confusion Matrix Analysis

The confusion matrices of the MLP, CNN, and RNN models are presented in Figure 5. The diagonal elements represent correctly classified samples, whereas the off-diagonal elements indicate misclassifications between flower categories.

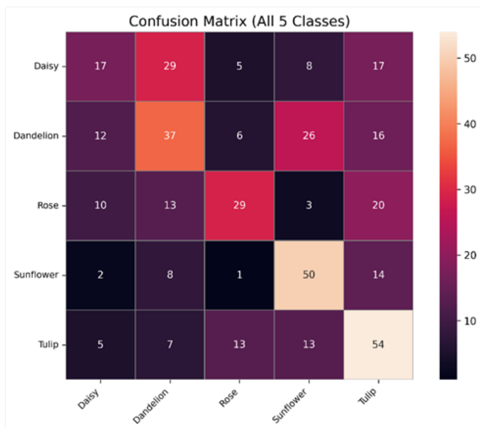
As shown in Figure 5, the CNN produced the strongest concentration of predictions along the main diagonal, indicating that the majority of test samples were correctly classified. Although some confusion remained between visually similar flower categories, the number of misclassified samples was substantially lower than that of the MLP and RNN.



(a) MLP



(b) CNN



(c) RNN

Figure 5: Confusion matrices of the evaluated architectures.

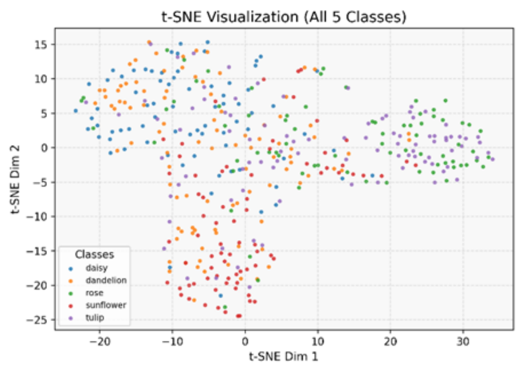
In contrast, both the MLP and RNN exhibited a greater number of off-diagonal entries, suggesting increased difficulty in distinguishing between similar flower classes. The MLP displayed the highest degree of class confusion, while the RNN achieved slightly better class discrimination but still produced considerably more misclassifications than the CNN.

Overall, the confusion matrices are consistent with the quantitative performance metrics presented earlier, further demonstrating that the CNN learned more discriminative spatial feature representations and achieved superior class discrimination on the flower image classification task.

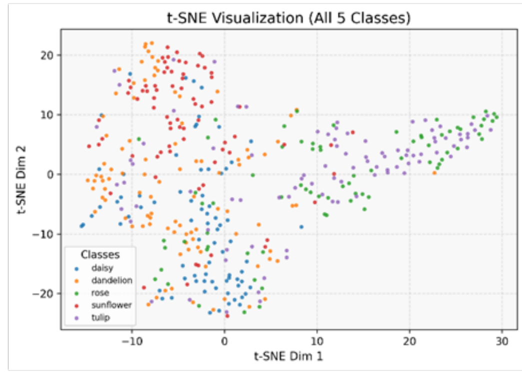
4.4 Feature Representation Analysis

The t-SNE visualizations of the learned feature representations for the CNN, MLP, and RNN models are shown in Figure 6. These visualizations project the high-dimensional feature embeddings into a two-dimensional space to illustrate how samples from different flower classes are organized.

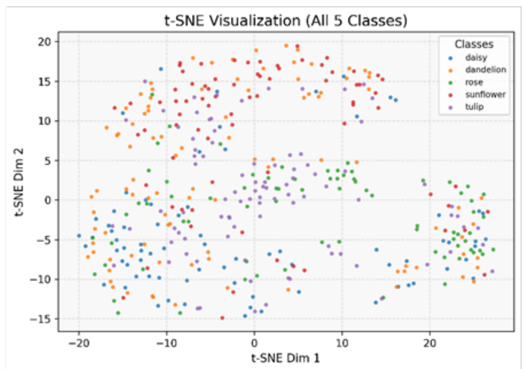
As shown in the figure, none of the three models produced a completely clear separation among the five flower categories. Considerable overlap exists between classes, with many samples concentrated in a central region and only a portion of the samples forming scattered clusters. This indicates that the learned feature representations are not fully discriminative, reflecting the visual similarity among several flower categories in the dataset.



(a) MLP



(b) CNN



(c) RNN

Figure 6: t-SNE visualizations of the learned feature representations.

Despite the overlapping distributions, the CNN exhibits relatively better organization of the feature space than the MLP and RNN, which is consistent with its superior classification performance. The MLP and RNN display greater mixing between classes, suggesting that they learned less distinctive feature representations under the same experimental conditions.

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4.5 Occlusion Sensitivity Analysis

Occlusion sensitivity maps of the CNN, MLP, and RNN models are presented in Figure 7. These visualizations illustrate how the predicted class confidence changes when different regions of an input image are occluded, thereby identifying the image regions that contribute most to each model’s classification decisions.

As shown in Figure 7, the CNN consistently exhibited broad and well-defined regions of importance across all five flower classes. The highlighted regions were primarily concentrated on the flowers themselves, indicating that the model relied on meaningful visual structures when making classification decisions.

In contrast, the MLP produced weaker and less consistent sensitivity patterns, with fewer clearly defined regions contributing to the final predictions. The RNN exhibited the weakest responses overall, showing only limited localization across most flower classes. Although the dandelion class displayed a more noticeable region of importance, the remaining classes generally exhibited diffuse or minimal responses compared with the CNN.

These qualitative observations are consistent with the quantitative evaluation results presented earlier. The CNN’s ability to consistently focus on informative regions of the input images aligns with its superior classification performance, whereas the weaker and less localized attention patterns observed for the MLP and RNN correspond to their comparatively lower classification accuracy.

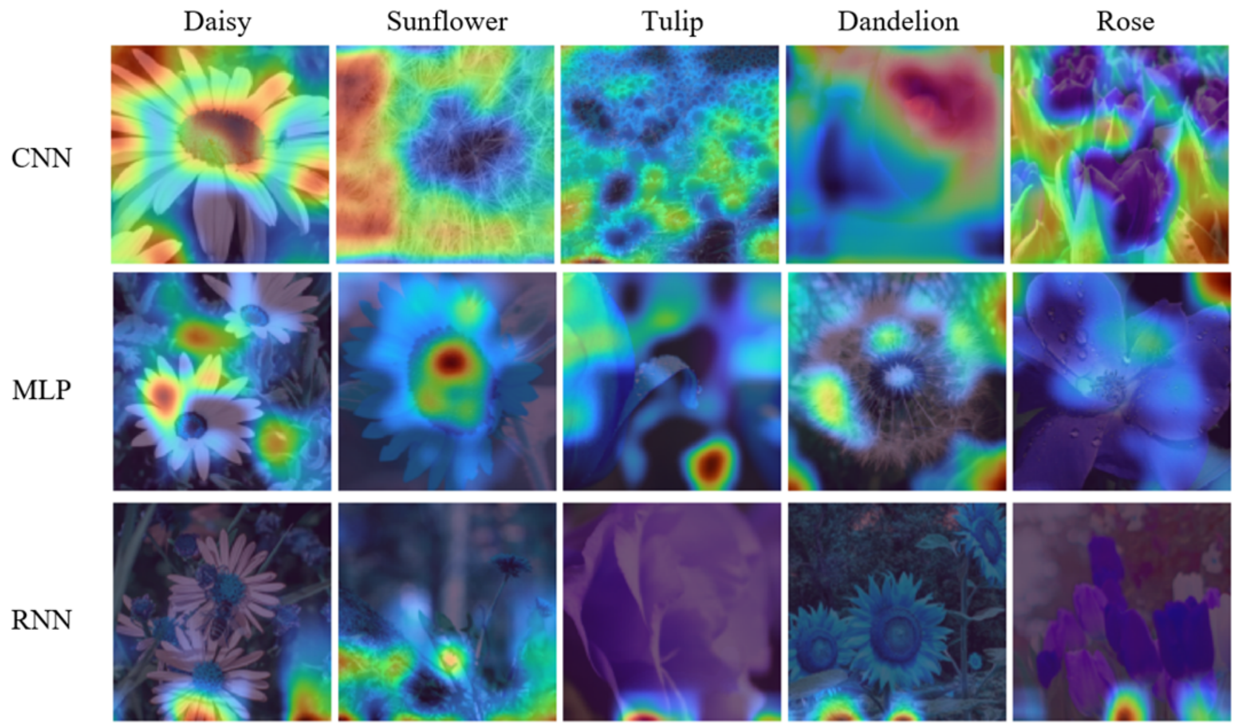

Figure 7: Occlusion sensitivity maps of the evaluated architectures. Rows correspond to the CNN, MLP, and RNN models, while columns correspond to the Daisy, Dandelion, Rose, Sunflower, and Tulip classes.

5 Discussion

5.1 Why Did the CNN Perform Better?

The CNN consistently achieved the best overall performance among the three evaluated architectures. This superiority can be primarily attributed to its architectural inductive bias, which is specifically designed for image data. Through local receptive fields and weight sharing, convolutional layers efficiently capture spatial patterns such as edges, textures, and object structures while preserving the relationships between neighboring pixels. As the network becomes deeper, these local features are progressively combined into higher-level semantic representations that are well suited for visual recognition tasks.

The experimental results presented in the previous section consistently support this explanation. Across both the quantitative evaluation metrics and qualitative visualizations, the CNN demonstrated stronger feature learning, better class discrimination, and more meaningful localization of informative image regions than the MLP and RNN. These findings indicate that the spatial inductive bias of convolutional neural networks provides a substantial advantage for image classification.

5.2 Why Did the MLP Perform Worse?

The MLP achieved lower classification performance because its architecture does not explicitly preserve the spatial structure of image data. Before entering the network, each image is flattened into a one-dimensional feature vector, causing the spatial relationships between neighboring pixels to be discarded. Consequently, the model must infer these relationships entirely from the training data, making the learning process less efficient.

This limitation is reflected in the experimental results. Compared with the CNN, the MLP learned less discriminative feature representations and exhibited weaker class separation during classification. These observations suggest that the absence of an image-specific inductive bias limits the MLP's ability to effectively model visual information under identical training conditions.

5.3 Why Did the RNN Perform Worse?

The RNN also performed worse than the CNN because recurrent neural networks are primarily designed for sequential data rather than two-dimensional images. In this study, each image was represented as a sequence of image rows and processed using an LSTM-based architecture. Although this representation enables the model to capture sequential dependencies, it does not naturally preserve the two-dimensional spatial relationships that are fundamental to image recognition.

The qualitative and quantitative analyses demonstrate that this sequential inductive bias is less appropriate for image classification than the spatial inductive bias of convolutional networks. Although the RNN achieved performance comparable to the MLP, it consistently learned less distinctive feature representations and produced lower classification performance than the CNN under the same experimental conditions.

5.4 Interpretation of the Experimental Results

The primary objective of this study was to investigate whether architectural inductive bias influences image classification performance when all other experimental factors are controlled. The results consistently support this hypothesis. Because all three models were trained using the same dataset, preprocessing pipeline, optimization strategy, hyperparameters, and evaluation protocol, the observed performance differences can be primarily attributed to their architectural characteristics.

Among the evaluated architectures, the CNN consistently produced the strongest quantitative and qualitative results. In contrast, the MLP and RNN exhibited similar performance levels but were less effective at learning discriminative visual representations. These findings demonstrate that matching a model's architectural inductive bias to the structure of the input data is a critical factor in achieving effective image classification performance. For image recognition tasks, architectures specifically designed to exploit spatial information provide a clear advantage over architectures intended for general vector processing or sequential data modeling.

5.5 Limitations of the Study

Several limitations should be considered when interpreting the findings of this study. First, the experiments were conducted using a single five-class flower image dataset. Consequently, the conclusions may not generalize to more diverse image classification tasks or larger benchmark datasets.

Second, relatively lightweight implementations of the CNN, MLP, and RNN architectures were adopted to ensure a fair comparison under identical experimental conditions. More advanced architectures or extensive hyperparameter optimization could potentially improve the performance of all three models.

Finally, this study focused on isolating the influence of architectural inductive bias rather than maximizing the performance of individual models. Therefore, the reported results should be interpreted as evidence of the effect of architectural design under a controlled experimental setting rather than as the highest achievable performance for each architecture.

6 Conclusion

This study investigated the influence of architectural inductive bias on image classification performance by comparing a Convolutional Neural Network (CNN), a Multi-Layer Perceptron (MLP), and a Recurrent Neural Network (RNN) under identical experimental conditions. All models were evaluated on the same five-class flower classification dataset using consistent preprocessing, training configurations, and evaluation criteria.

The results demonstrated that the CNN consistently achieved superior performance compared with the MLP and RNN across quantitative metrics. The qualitative analyses further showed that the CNN produced fewer classification errors, learned better-organized feature representations, and focused on more informative image regions during prediction. In contrast, the MLP and RNN showed weaker performance due to their limited ability to exploit spatial information in images.

Overall, the findings confirm that architectural inductive bias significantly influences image classification performance. Models designed with image-specific spatial priors, such as CNNs, are better suited for visual recognition tasks than architectures based on general vector processing or sequential modeling.

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